

PHIET TRAN

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EDUCATION

University of California, Los Angeles (UCLA)

Expected June 2028

Bachelor of Science in Mathematics and Computer Science, Minor in Data Science Engineering

Los Angeles, California

- Relevant Coursework: Data Structures, Algorithms & Analysis, Software Construction, Operating Systems, Fundamentals of Artificial Intelligence, Linear Algebra, Real Analysis, Differential Equations

RELEVANT EXPERIENCE

SONNY (Shop Online New York)

February 2026 – Present

Software Engineer Consultant - Import Tooling and Image Uploading

Los Angeles, California

- Built import tooling and image-handling features for a live e-commerce admin platform on a **Next.js/Spring Boot** stack, coordinating delivery with a 7-person engineering team under a compressed timeline
- Designed field-mapping logic for a CSV/Excel-to-product import pipeline, transforming spreadsheet rows into structured product data across 6 fields to enable automated bulk product creation and replace manual single-product entry
- Integrated the pipeline with a 20+ endpoint **Java/Spring Boot REST API** to support automated bulk product creation
- Extended the Cloudinary-backed image upload system beyond its 4-image, 5MB-per-file cap to support bulk multi-image upload, building drag-and-drop reordering and primary-image selection in **Next.js/TypeScript**

DataRes @ UCLA

September 2025 – Present

Data Science Project Lead - YouTube Data Analysis

Los Angeles, California

- Engineered a data cleaning pipeline reducing a 100,000-video YouTube dataset to 1,166 validated records using engagement and watch-time validity filters in **Python** (pandas)
- Led statistical analysis on screen time, sleep quality, and stress; developed KDE plots, correlation heatmaps, and 3D scatter plot using seaborn and matplotlib to surface relationships with digital media and mental health
- Directed data analysis with team of 4 from dataset processing through publication, leading both analysis coordination and article writing; findings published to 1,000+ readers on DataRes Medium blog

TECHNICAL PROJECTS

Dormi: IoT Sleep Environment Monitor | C++, TypeScript, React, Node.js, SQLite

- Built a **React/TypeScript** dashboard displaying real-time CO, temperature, humidity, and light sensor readings with 24-hour historical trends, sleep scores, interactive visualizations, and 60-second auto-refresh
- Developed an **Express/Node.js REST API** with **SQLite** to ingest, store, and retrieve real-time and historical environmental sensor data for dashboard visualization and sleep-score computation
- Designed a research-backed 0–100 sleep scoring algorithm using nonlinear penalty functions for CO, temperature, humidity, and light, including temperature-humidity interaction effects
- Integrated OpenAI's GPT-4o-mini to generate personalized sleep recommendations by querying 7 nights of aggregated sensor history and supplying environmental measurements and sleep thresholds as context

StudySearcher | React, Node.js, Express, PostgreSQL, Socket.IO

- Architected a full-stack PERN study-partner platform with 8 relational Postgres tables and 17 REST endpoints across 8 route modules, covering auth, profiles, matching, messaging, and moderation
- Designed **SQL** interval-overlap matching algorithm pairing students by shared availability
- Shipped transactional bidirectional user-blocking with cascading conversation deletion, validated via Playwright E2E tests
- Engineered conversation-open API with guard-clause validation, block-list checks, and deduplication join across membership records to prevent duplicate 1:1 chats, enforcing least-privilege access

Real-Time Waste Detection System | JavaScript, TensorFlow.js, HTML/CSS

- Built a real-time waste detection web application using JavaScript, HTML, CSS, and TensorFlow.js, integrating live camera input with a computer vision model achieving 92% accuracy across 5 waste categories
- Engineered a responsive interactive interface with real-time classification, dynamic point tracking, and competitive leaderboards, integrating a model trained on 5,000+ labeled waste images

TECHNICAL SKILLS

Programming Languages: C/C++, Java, JavaScript/TypeScript, Python, SQL, HTML/CSS, Bash/Shell

Frameworks/Libraries: React, Next.js, Node.js, Express, Spring Boot, Socket.IO, Pandas, NumPy, PyTorch, TensorFlow

Databases: PostgreSQL, MySQL, MongoDB, SQLite

Developer Tools: Git, Docker, Playwright, Cloudinary, Jupyter